Architecting a Human-Interpretable, Hyper-Fast Generative Decoder via Vector Symbolic Architectures and Hypernetworks

The contemporary trajectory of large language models and generative decoders is fundamentally constrained by a severe, compounding dual crisis: an unsustainable computational hardware bottleneck and an intrinsically opaque, black-box operational nature. The foundational reliance on dense floating-point matrix multiplications, autoregressive token generation, and global softmax attention mechanisms renders state-of-the-art architectures prohibitively expensive, dictating a reliance on massive multi-GPU clusters and vast High-Bandwidth Memory capacities.[1] Simultaneously, the dense continuous vector spaces defining these conventional models resist transparent mechanistic interpretability, deeply obscuring the internal logic, feature representations, and causal pathways dictating their outputs.[2] To construct a generative decoder that is both "Hyper Fast"—capable of operating on strict 12GB VRAM consumer hardware with minimal latency—and intrinsically human-interpretable, a radical paradigm shift in both data representation and algorithmic execution is unconditionally required.[1]

This comprehensive analysis formalizes the theoretical, mathematical, and practical framework for a neurogenesis-driven, hyperdimensional generative decoder. By synthesizing the algebraic rigor of Vector Symbolic Architectures, the structured combinatorial priors of Geometric Group Theory, the high-throughput parallel efficiency of Hypernetwork Linear Attention, and the parallel execution mechanics of Hyper-Non-Autoregressive decoding, a novel architectural standard is established.[1] Furthermore, by incorporating Sanger's Generalized Hebbian Algorithm for online whitening to preserve Signal-Signal Factorization, and deploying the system via a vendor-agnostic Vulkan Compute pipeline, the resulting architecture achieves unprecedented processing velocity while maintaining mathematically guaranteed structural transparency.[1]

The Holographic Paradigm: Vector Symbolic Architectures

The epistemological departure from standard von Neumann computational models and traditional localized neural network architectures lies in the adoption of Hyperdimensional Computing, operating under the broader algebraic framework of Vector Symbolic Architectures.[1] Instead of utilizing precise, localized scalar values or dense continuous floating-point tensors, information is distributed holistically across massive, fixed-length

hypervectors.[1] These hypervectors typically operate in high-dimensional spaces ranging from a minimum of one thousand dimensions to upwards of ten thousand dimensions.[1]

In a binary or bipolar space where vectors exist in the domain of either bipolar or binary states, the mathematical geometry of the high-dimensional space dictates that any two randomly drawn hypervectors are nearly orthogonal.[1] Consequently, their expected cosine similarity or fractional dot product approaches zero.[1] This pseudo-orthogonality allows for a highly robust, holographic representation.[1] Because information is uniformly distributed across the entire hypervector rather than isolated in specific units, the degradation, precision loss, or corruption of individual vector components does not result in catastrophic memory loss.[1] This intrinsic robustness heavily favors the aggressive quantization and bit-packing strategies required for memory-constrained hardware deployments.[1]

Information processing within this paradigm is executed via three primary algebraic operations, which together form a complete, ring-like mathematical structure over the high-dimensional vector space.[1] The deterministic manipulation of these symbols using mathematically reversible operations provides the foundational bedrock for mechanistic interpretability.[5] Unlike deep continuous networks where the activation of a hidden unit is arbitrarily derived from stochastic gradient descent, these geometric bindings create transparent hierarchical linkages.[1]

VSA Operation	Mathematical Implementation	Algorithmic Function	Impact on Vector Capacity
Binding	Element-wise multiplication or bitwise XOR	Associates variables with values; creates hierarchical data structures and temporal linkages.	Generates a new vector mathematically orthogonal to operands; neutral to superposition limits.
Permutation	Cyclic coordinate shift	Encodes sequential order, temporal dynamics, and positional embeddings without commutative ambiguity.	Preserves orthogonality and total dimensionality; completely neutral to superposition limits.

Bundling	Element-wise addition followed by thresholding	Combines multiple vectors into a composite set memory; enables associative nearest-neighbor retrieval.	Reduces Signal-to-Noise Ratio (SNR); actively drives the system toward capacity saturation and crosstalk.

The capacity to unbind these vectors relies heavily on the specific encoding regime utilized during the initial mapping phase. The mapping of continuous or sequential data into the hyperdimensional space must be carefully tuned depending on the explicit operational goal of the model, forcing a choice between learning optimization and cognitive retrieval.[7] In kernel-based hyperdimensional encoders, the control mechanism for navigating this transition is the kernel width.[7]

Machine learning classification tasks traditionally benefit from correlative encodings.[7] Correlated representations capture shared structures and overlapping statistical patterns across different samples.[7] By widening the kernel width, the encoder maximizes memorization and generalization capacity, allowing the model to adapt flexibly to novel inputs.[7] In this regime, adding more training samples enhances learning capacity, with optimal distribution separation values hovering between zero and slightly above one.[7]

Conversely, cognitive tasks such as precise question answering, symbolic reasoning, and human-interpretable decoding demand exclusive encodings.[7] Narrowing the kernel width maximizes separation between data points, creating highly separable, strictly orthogonal hypervectors.[7] This exclusivity is an absolute requirement for accurate information retrieval, as it prevents error propagation during complex factorization and decoding operations.[7] A human-interpretable decoder mandates this exclusive encoding regime, requiring separation values of approximately two to three to guarantee that bundled concepts can be distinctly separated, unbound, and analyzed without catastrophic interference.[7]

Resolving the Autoregressive Bottleneck via Hyper-NAR Decoding

Standard autoregressive language models are inherently sequential; generating a specific token strictly requires the full computed context of all previous outputs within the sequence.[1] This sequential dependency dictates that generation latency scales linearly with sequence length, fundamentally limiting throughput and exacerbating the memory bandwidth bottleneck as context windows expand.[1] Standard Non-Autoregressive Transformers attempt to solve this computational inefficiency by predicting parallel blocks of text simultaneously.[1] However, these

traditional non-autoregressive models notoriously suffer from conditional total correlation loss.[1] By predicting tokens independently in parallel, they fail to capture the deep grammatical and semantic dependencies between adjacent tokens, frequently resulting in logically incoherent, hallucinated, or highly repetitive textual outputs.[1]

The integration of Vector Symbolic Architectures and dynamic hypernetworks enables the Hyper-Non-Autoregressive decoding mechanism, elegantly circumventing both the latency of autoregressive models and the correlation loss of standard parallel decoders.[1] In the Hyper-Non-Autoregressive framework, the decoder does not probabilistically guess individual discrete text tokens.[1] Instead, the hypernetwork is tasked with predicting a single transformation hypervector.[1] This generated hypervector mathematically represents the exact algebraic operation required to transition the system from its current holistic state directly to the targeted output state.[1]

The structural transition is enacted by directly applying the binding operation to the current state vector and the predicted transformation hypervector.[1] Because the latent space is grounded in a deterministic role-filler binding scheme, the resulting composite vector instantly encapsulates the updated global semantic structure.[1] To extract readable output tokens from this generated hypervector, the architecture leverages a highly specialized retrieval process.

Because bundled and transformed hypervectors represent a noisy approximation of the exact target concepts due to the superposition of multiple elements, they must be passed through a high-speed clean-up memory.[1] By operating entirely in the strict bipolar domain, the architecture abandons computationally expensive floating-point cosine similarity metrics.[1] Instead, the system leverages Facebook AI Similarity Search configured with an optimized binary index structure.[1] This allows the system to compare the generated semantic signature against a static dictionary of valid atomic hypervectors using native hardware bitwise operations, specifically XOR and hardware-accelerated population counts.[1]

This algorithmic transition from dense continuous matrix multiplication to bitwise Hamming distance computation drastically reduces memory bandwidth requirements.[1] Billion-scale vector retrievals are executed in sub-millisecond timeframes, transforming the inference bottleneck from an optimization-heavy computational burden into an ultra-fast, highly optimized database routing problem.[1] Furthermore, to resolve the conditional dependency problem inherent to traditional non-autoregressive decoding, the module generates latent conditioning codes for the hypernetwork based on the holistic state.[1] The hypernetwork then dynamically instantiates a highly specific, lightweight decoder sub-network optimized purely for that exact context.[1] Because this dynamic sub-network relies on the inherent structural preservation of the hyperdimensional space, the generated decoder implicitly retains the internal logical dependencies and left-to-right conditional knowledge that standard parallel decoders lose, achieving a "Plan-then-Generate" execution framework.[1]

Decoding Paradigm	Execution Methodology	Inference Latency	Dependency Modeling	Primary Retrieval Metric	Generation Basis
Standard Autoregressive	Sequential, Token-by-Token	High, scales linearly	Maintained via sequential feedback	Softmax-driven Cosine Similarity	Probabilistic token sampling
Standard Non-Autoregressive	Parallel block prediction	Low	Poor, suffers correlation loss	Softmax-driven Cosine Similarity	Probabilistic token sampling
Hyper-NAR with VSA	Parallel decoding of dynamic weights	Very Low, sub-millisecond	Strong, encoded via hypervector prior	Bitwise Hamming Distance	Deterministic geometric trajectory

The Geometry of Mechanistic Interpretability

While high-throughput sequence generation is critical for consumer hardware deployments, the primary theoretical advancement of this architecture is its absolute, mathematical interpretability.[2] Unlike dense continuous neural networks where the activation of a hidden unit lacks simple human-understandable meaning, the hyperdimensional paradigm allows for precise, mechanistic analysis of internal states without relying on post-hoc black-box approximations.[10]

Standard transformer embeddings utilize cosine similarity as a proxy for statistical co-occurrence, a purely correlative metric with no underlying logical structure.[1] The Hyper-Non-Autoregressive decoder, however, explicitly utilizes group-theoretical structures as an inductive prior to enforce strict combinatorial generalization.[1] The encoding function employs a deterministic role-filler binding scheme modeled mathematically after the combinatorial mechanics of the Rubik's Cube permutation group.[1]

Specific positions within the data sequence are represented by deterministic position vectors acting as structural roles, analogous to the fifty-four discrete spatial positions of a standard Rubik's Cube.[1] Filler vectors representing the actual semantic data features are bound to these structural roles using the bitwise binding operation.[1] To ensure that the bundled

representations remain strictly bipolar and prevent unbounded value drift, a signum thresholding mechanism is applied.[1] This snapping mechanism preserves the core algebraic properties of the space, ensuring that the binding operation remains perfectly self-inverse.[1]

Because the latent semantic space is mathematically grounded in a permutation group, operations within this space correspond to disjoint cycles applied simultaneously to the vector array.[1] These transformations rigorously satisfy group theory identity laws; for example, applying four consecutive identical rotational transformations mathematically guarantees a return to the initial state.[1] The profound implication for mechanistic interpretability is that structural dissimilarity in the data correlates directly with mathematical vector distance.[1]

In this strictly structured space, the Hamming distance between two hypervectors correlates mathematically with God's distance, defined as the absolute minimum number of algorithmic operations required to transition from one structural state to another.[1] Consequently, sequence generation is no longer a process of stochastic, probabilistic token prediction; it is a deterministic, highly calculated trajectory through a structurally preserved combinatorial graph.[1] Every computational step is mathematically bounded, rendering the generation path fully traceable.[1]

The Hyperdimensional Probe and VSA Unbinding

To extract human-readable interpretability during the decoding phase, the system directly employs the unbinding operation.[13] Because binding in a binary space via the XOR operator is its own inverse, the unbinding operation can mathematically factor out specific constituents from a highly composite hypervector.[3] If a composite hypervector represents a complex semantic state created by binding a specific concept with a specific grammatical or structural role, applying the unbinding operation with the known role exactly isolates the target concept.[5] This allows researchers and automated interpretability systems to extract explicit logical rules, analogical relationships, and distinct semantic features directly from the model's internal memory without destructive interference.[14]

This analytical capability is formalized operationally as the Hyperdimensional Probe.[3] Current mechanistic interpretability methods traditionally focus either on input-oriented feature extraction or output distribution inspection.[3] Methods like Sparse Autoencoders attempt to learn a sparse proxy representation but frequently suffer from feature explosion, generating millions of distinct features whose naming often suffers from vagueness, verbosity, and extreme data dependence.[10] Conversely, direct logit attribution focuses only on the final output distribution, entirely ignoring the intermediate logic.[3]

The Hyperdimensional Probe unifies these fractured perspectives by mapping the generative language model's residual stream into a controlled proxy space explicitly structured by hyperdimensional encodings.[3] Functioning as a hybrid supervised probe, it extracts

input-aligned latent features while fully supporting output-oriented interpretability.[3] If the generative decoder makes an erroneous prediction or hallucinates a concept, analysts can apply the unbinding operation to the intermediate sequence of hypervectors to trace precisely which semantic concept or structural role was corrupted during the deterministic trajectory.[3] This fundamentally resolves the black-box alignment problem, providing deep semantic understanding of neural representations.[3]

Signal-Signal Factorization and the Capacity Crisis

The capability to seamlessly unbind and interpret these vectors relies entirely on maintaining a specific geometric property known as Signal-Signal Factorization.[1] Signal-Signal Factorization, synonymous in this context with disentanglement, occurs only when independent environmental variables or semantic concepts are encoded along distinct and orthogonal directions of variation within the neural activity space.[1]

Generalization performance and interpretability in neural networks are determined by four critical geometric statistics: Signal-Signal Factorization, Signal-Noise Factorization, total dimensionality, and correlation structures.[1] In an ideal, mathematically perfect system, every newly bound concept occupies a perfectly orthogonal subspace.[1] However, because real-world linguistic and sequential data contains highly correlated features, the continuous bundling of vectors creates a composite representation that accumulates massive variance.[1] As the number of bundled vectors increases, the variance expands, and the relative mathematical distance between the composite vector and any individual constituent shrinks.[1]

This phenomenon manifests as an accumulation of severe crosstalk, acting as additive noise during the nearest-neighbor retrieval process.[1] At a specific mathematical juncture, the Signal-to-Noise Ratio falls below the critical margin of error required for accurate unbinding.[1] When the system bundles data up to its heuristic 75 percent capacity threshold, the pseudo-orthogonality completely breaks down.[1] Without maintaining Signal-Signal Factorization, the representation violently entangles, collapsing into a dense, uninterpretable state where original constituents can no longer be reliably distinguished from surrounding noise.[1] This collapse triggers catastrophic forgetting, rendering the decoder useless.[1] To maintain long-term interpretability, the generative decoder requires an active, online mechanism to combat this representational collapse.

Online Whitening and Algorithmic Neurogenesis

To architect an adaptive, lifelong-learning generative decoder that successfully circumvents capacity saturation and maintains mathematically rigorous interpretability, the architecture implements dynamic structural expansion.[1] Adhering to the biological sparse updating hypothesis, the system maintains a static, highly efficient dimensionality until the exact 75 percent threshold is breached.[1] When the bundled hypervector crosses this boundary, the

system halts standard associative bundling and triggers a sparse update, executing a computational process analogous to biological adult neurogenesis.[1]

Before new hypervector dimensions can be added to the architecture, the saturated, highly correlated representation must be systematically disentangled to restore Signal-Signal Factorization.[1] Simply appending new random dimensions to a vector already suffering from severe crosstalk does not resolve the underlying interference.[1] This resolution is achieved through a linear transformation known as whitening, or sphering.[1] Whitening converts a vector of random variables with a known covariance matrix into a set of new variables whose covariance is the identity matrix.[1]

By mathematically flattening the eigenvalue spectrum of the data covariance matrix, whitening ensures that signal variance is distributed equally across all dimensions.[1] This effectively decorrelates overlapping linguistic patterns, compresses redundant associative data, and strictly separates the true semantic signal from interference noise, thereby enhancing Signal-Noise Factorization.[1] In complex perception models, this precise removal of chance correlations provides an extraordinary improvement in sampling efficiency and estimation accuracy.[1] Most critically for the human-interpretable decoder, whitening extracts the true, independent latent features of the text, rendering the previously entangled representation fully factorized and interpretable once more.[1]

Sanger's Generalized Hebbian Algorithm

Standard mathematical whitening presents a severe computational problem; it requires computing the full covariance matrix of the data stream and performing an offline Eigenvalue Decomposition or Principal Component Analysis.[1] This mathematical operation carries a prohibitively heavy computational complexity of $O(N^3)$.[1] Imposing an operation with cubic complexity would instantly stall the high-speed inference pipeline, entirely negating the benefits of the hyperdimensional architecture.[1]

To completely circumvent this computational bottleneck, the decoder utilizes Sanger's Generalized Hebbian Algorithm.[1] Sanger's rule is a biologically plausible, unsupervised Hebbian learning mechanism that serves as a multi-component expansion of Oja's rule.[1] While Oja's rule introduces a homeostatic decay mechanism through subtractive normalization to discover only a single principal component representing the direction of maximum variance, Sanger's algorithm incorporates a Gram-Schmidt-like orthogonalization process directly into the localized synaptic update.[1]

For the i-th neuron or feature dimension in the network, the weight update mathematically subtracts the projections of the incoming input onto all previously learned $i-1$ principal

components.[1] The multi-unit update formula dictating this extraction is formally defined as:

$$\Delta w_{ij} = \eta y_i \left(x_j - \sum_{k-1}^{i} y_k w_{kj} \right)$$

By forcing each subsequent vector dimension to learn strictly from the residual variance left over by the preceding dimensions, Sanger's algorithm guarantees that the resulting weight vectors become perfectly orthogonal to one another.[1] This algorithmic breakthrough allows the decoder to extract multiple orthogonal components sequentially in an online, streaming fashion with a linear computational complexity of $O(N)$, entirely without explicitly computing a covariance matrix.[1]

Following the completion of the whitening phase via Sanger's algorithm, the system isolates the residual error vector, calculated precisely from the difference between the saturated input and its whitened reconstruction.[1] This residual error is mathematically guaranteed to be strictly orthogonal to the pre-existing, saturated space.[1] The architecture utilizes this orthogonal residual error to initialize a newly allocated memory block, seamlessly expanding the hypervector dimensionality.[1] This algorithmic neurogenesis immediately restores the system's Signal-to-Noise Ratio to absolute baseline, completely bypassing the 75 percent capacity limit while perfectly preserving historical data traces.[1] The newly disentangled, interpretable features now reside securely in distinct, orthogonal vector dimensions, securing long-term Signal-Signal Factorization.[1]

Hyper Fast Execution via Vulkan Compute

The theoretical advancements of deterministic hypervector trajectories and continuous algorithmic neurogenesis must be paired with extreme low-level systems engineering to achieve truly Hyper Fast generation on standard edge hardware constrained to 12GB of Video RAM.[1] This is accomplished by replacing the standard multi-head attention mechanism with Hypernetwork Linear Attention and executing the entire mathematical pipeline via highly optimized, bit-packed, vendor-agnostic Vulkan Compute shaders.[1]

Standard Multi-Head Attention maps sequential inputs into queries, keys, and values to compute an attention matrix, an operation strictly dependent on a global softmax function.[1] Softmax introduces a massive computational bottleneck because it demands costly synchronization barriers across the GPU, forcibly halting parallel execution to compute global probabilities.[1] The Hypernetwork Linear Attention framework entirely recasts this standard attention mechanism as a shared hypernetwork.[1] In this architecture, low-dimensional latent codes derived from attention scores dynamically configure the weights of a layer-local value network.[1]

Crucially, the architecture modifies this generated value network from a linear to a non-linear operation by introducing a local nonlinearity directly into the aggregation step.[1] When mapped into the Binary Hyperdimensional Transformer domain, this local nonlinearity maps flawlessly to the thresholding function inherent to the bundling step of Vector Symbolic Architectures.[1] Because this thresholding operation evaluates inputs locally based on the query and key index, it requires absolutely no communication across separate keys.[1] This mathematical substitution entirely eliminates the global softmax bottleneck, immediately unlocking unimpeded, hardware-saturating parallel execution across the entire generative decoder.[1]

Hardware Compression and Subgroup Reductions

To successfully fit an ever-expanding, lifelong-learning sequence model within the strict 12GB VRAM threshold, the architecture eschews dense 32-bit floating-point tensors in favor of highly compressed bit-packed binary hypervectors.[1] In a traditional deep learning model, a single continuous vector operating in ten thousand dimensions utilizing standard 32-bit floats consumes forty kilobytes of VRAM.[1] Scaling this quadratically for a standard eight-thousand token context window instantly triggers fatal Out-of-Memory cascades on consumer hardware.[1]

By mapping the model exclusively into the binary domain, a ten thousand-dimensional binary hypervector is densely packed into 32-bit unsigned integers.[1] This reduces the memory requirement to a mere 313 integers, translating to approximately 1.25 kilobytes per vector.[1] This represents a massive 32x memory compression ratio, mathematically guaranteeing that the 12GB limit is vastly sufficient to house the entire sequence model, even as continuous algorithmic neurogenesis dynamically scales the total dimensionality throughout the model's operational lifespan.[1]

Deploying this highly compressed architecture across disparate consumer hardware ecosystems requires a compute API capable of exacting low-level hardware control.[1] The Vulkan Compute API is uniquely suited due to its explicit memory management and mandatory cross-platform support across diverse silicon.[1] The computational core of the attention calculation requires executing a Hamming similarity matrix using bitwise XOR operations, immediately followed by a massive parallel reduction across thousands of dimensions.[1] Naive implementations utilizing atomic counter loops heavily bottleneck memory bandwidth.[1]

To achieve maximum execution velocity, the architecture exploits the Vulkan shader subgroup arithmetic extension.[1] Modern GPUs execute computational threads in parallel lock-step groupings known as subgroups, warps, or wavefronts.[1] The compute shader dynamically queries the physical device's subgroup size at runtime to maintain absolute vendor-agnosticism.[1] By bit-packing the arrays, a single thread processes thirty-two vector dimensions simultaneously.[1] The shader computes the local XOR bit count and immediately

utilizes the hardware intrinsic subgroup addition function.[1]

This operation performs a lightning-fast parallel reduction across all invocations in the subgroup in a single clock cycle.[1] On architectures operating with 32-thread subgroups, this instantly aggregates 1,024 dimensions per instruction; on architectures utilizing 64-thread subgroups, it seamlessly scales to aggregate 2,048 dimensions per instruction without any branching divergence or code modification.[1] This highly optimized parallel reduction perfectly saturates the Arithmetic Logic Units of all major hardware architectures equally, ensuring maximum theoretical throughput.[1]

Dynamic Memory Scaling via Buffer Device Address

The implementation of continuous algorithmic neurogenesis poses a severe structural challenge: it requires allocating new hypervector dimensions at runtime without halting or stalling the high-speed inference pipeline.[1] Traditional GPU memory management relies heavily on static Descriptor Sets and Uniform Buffer Objects.[1] These rigid structures require pre-allocated, fixed-size memory bindings that are exceedingly cumbersome to resize during active execution.[1] Attempting to preemptively allocate massive contiguous blocks of memory to account for theoretical future vector growth inevitably triggers fatal memory errors on strictly constrained hardware.[1]

To achieve highly efficient, lazy capacity growth that accommodates neurogenesis flawlessly, the architecture strictly mandates the use of the Vulkan Buffer Device Address extension.[1] This extension exposes genuine GPU virtual addresses directly to the host application as standard 64-bit integers, allowing them to be manipulated as raw memory pointers directly within the compute shader.[1]

When the 75 percent capacity threshold is breached and the online whitening phase completes, the host application dynamically allocates a new, distinct block of device-local memory specifically reserved for the newly generated orthogonal dimensions.[1] Instead of pausing the GPU execution queue to reconstruct and bind massive new Descriptor Sets, the host application simply updates a 64-bit pointer located within a highly lightweight Push Constant array.[1] On the very next computational dispatch, the shader instantly dereferences this new pointer, accessing the expanded hypervector space seamlessly and instantaneously.[1] This advanced memory methodology entirely circumvents VRAM overallocation limits, eliminates descriptor binding overhead, and enables the real-time construction of dynamically expanding, interpretable neural architectures directly within the GPU memory space.[1]

Conclusion

The convergence of Vector Symbolic Architectures, Hypernetwork Linear Attention, and Vulkan-accelerated non-autoregressive decoding establishes a highly robust framework capable of solving the dual crises of computational latency and black-box opacity defining

modern language models. By mapping generative sequence modeling into a strict bipolar vector space, the architecture successfully abandons the opaque complexity of continuous floating-point networks. Utilizing the structural priors of Geometric Group Theory, sequence generation transitions from probabilistic, stochastic token guessing into a highly deterministic, mathematically bounded trajectory where Hamming distance serves as an exact, quantifiable metric of structural transformation.

The integration of the Hyperdimensional Probe and the unbinding operation ensures that all semantic logic, role-filler relationships, and feature representations remain intrinsically human-interpretable throughout the execution cycle. To prevent the collapse of Signal-Signal Factorization and maintain this interpretability under heavy cognitive loads, the architecture implements a biologically inspired capacity threshold. When saturation occurs, Sanger's Generalized Hebbian Algorithm performs continuous, online whitening, decorrelating entangled representations and extracting true independent latent features to drive algorithmic neurogenesis without stalling inference. By executing this comprehensive framework through a low-level, bit-packed Vulkan Compute pipeline—leveraging vendor-agnostic subgroup reductions and virtual pointer memory scaling—the architecture entirely bypasses traditional autoregressive bottlenecks, yielding an exceptionally fast, fully interpretable generative decoder optimized for deployment on highly constrained edge hardware.

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